

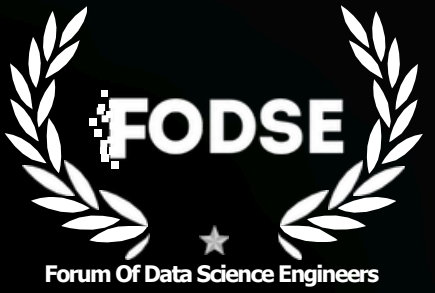


TEAM NAME : Singularity

COLLEGE NAME : SVCE

PROBLEM STATEMENT : Enhance Urban Living Through AI-Driven Traffic Management System

PROBLEM STATEMENT

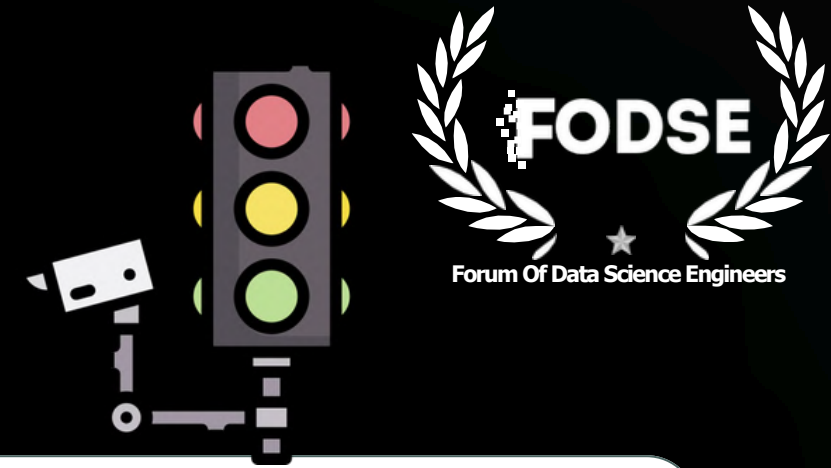


AI-DRIVEN TRAFFIC MANAGEMENT SYSTEM

Traffic congestion is a major challenge in urban areas, leading to delays, fuel wastage, and increased pollution. The goal is to develop a software-based intelligent traffic management system that uses AI to analyze real-time and historical traffic data to predict congestion and optimize signal timing strategies. The system should provide live traffic insights through a web or mobile dashboard and recommend alternative routes to improve traffic flow and reduce travel time.

SOLUTION

AI-DRIVEN TRAFFIC MANAGEMENT SYSTEM



ACCIDENT & PEDESTRIAN SAFETY

AI anticipates accidents and pedestrians to improve road safety and prevent traffic disruptions.

REAL-TIME TRAFFIC MONITORING

AI camera continuously monitor vehicle flow, speed, and lane density in real time.

AI TRAFFIC ANALYSIS

Reinforcement models analyze traffic patterns to detect congestion and abnormal vehicle behavior.

DYNAMIC SIGNAL CONTROL

Traffic signals automatically adjust green and red timings based on real-time traffic density.

EMERGENCY VEHICLE PRIORITY

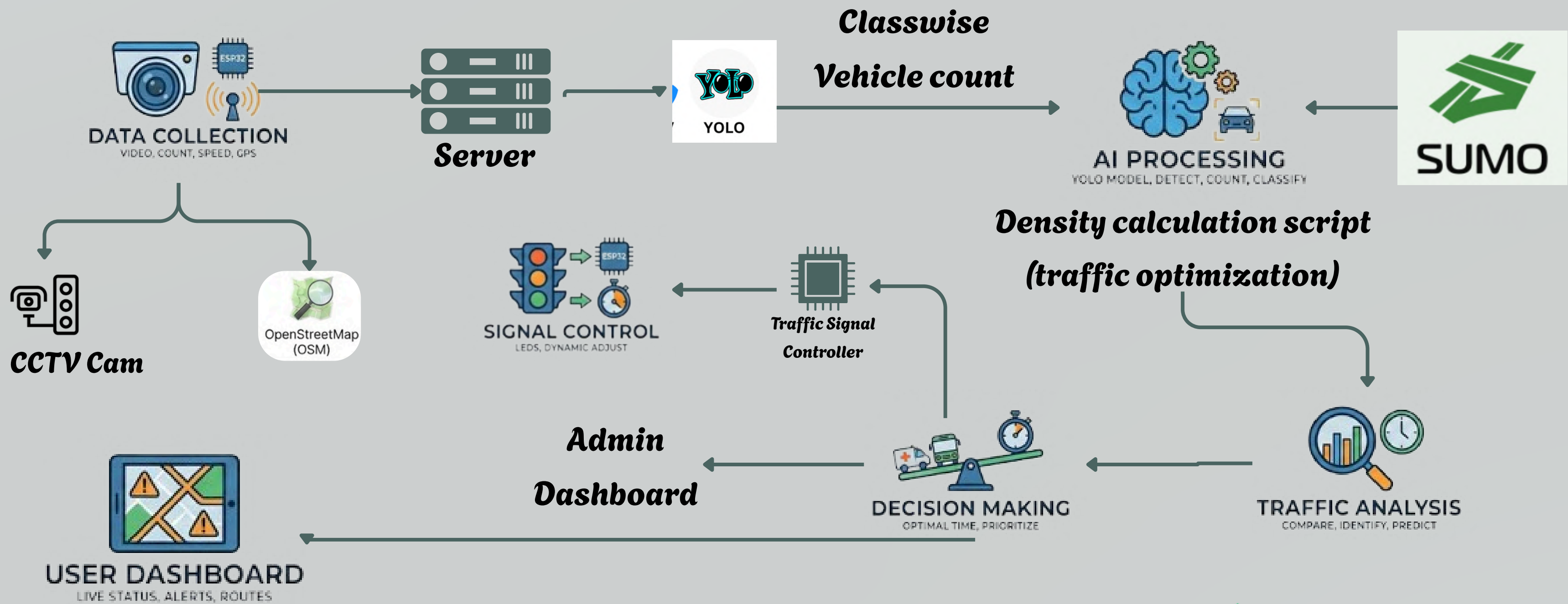
The system detects ambulances and fire trucks and creates a green corridor for fast emergency response.

SMART DASHBOARD & ROUTE GUIDANCE

Live dashboard displays traffic status, congestion, and optimized travel routes.



FLOW DIAGRAM



TECHNICAL REQUIREMENTS

FRONTEND:- React.js (interactive dashboards & real-time UI)

BACKEND:- Node.js (APIs & server logic)

DATABASE:- PostgreSQL (structured traffic data)

ML Model:- YOLO (detect vehicles & pedestrians etc)

Computer Vision:- OpenCV (process camera feeds)

Traffic Simulation:- SUMO (simulate traffic flow & signals)

Programming Languages:- Python, JavaScript (AI, CV, frontend/backend)

NOVELTY / USE CASE

AI-BASED ADAPTIVE SIGNAL CONTROL

Traffic signals dynamically adjust based on real-time vehicle density for optimized traffic flow

EMERGENCY VEHICLE TRACKING

Accurately tracks emergency vehicles gives priority to ambulances or emergency vehicles.

SMART ROUTE RECOMMENDATION

Uses historical and real-time data to forecast congestion and suggest alternate routes.

USE CASE

- ***Real-Time Traffic Monitoring***

Enables cities to monitor traffic in real-time, respond quickly to accidents, and manage congestion efficiently.

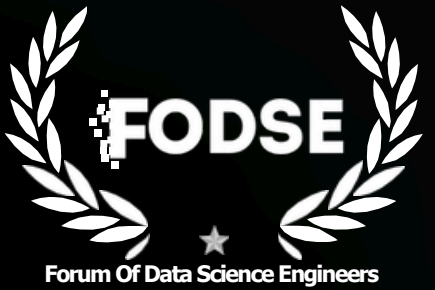
- ***Traffic Simulation & Urban Planning***

Helps urban planners test new road designs and traffic strategies before implementation, reducing costly mistakes

- ***Emergency Vehicle Prioritization***

Ensures emergency vehicles reach their destination faster by clearing traffic and providing uninterrupted routes.

BUSINESS MODEL / TARGET MARKET



BUSINESS MODEL

B2G-BUSINESS TO GOVERNMENT

The government (National Highway Authorities or Municipal Corporations) is our target Client.

PPP-PRIVATE PUBLIC PARTNERSHIP

Collaboration with government agencies to share resources and leverage expertise for large-scale deployments.

CCTV-COST MODEL

Cisco is a major player in the CCTV camera tenders, utilising these infrastructure for the camera feed.

TARGET MARKET

- **Government & Public Sector**

Traffic Departments / Transport Authorities: For real-time traffic monitoring, accident detection, and congestion analysis.

- **Private Sector**

Logistics and Delivery Companies: Optimize delivery routes using traffic predictions.

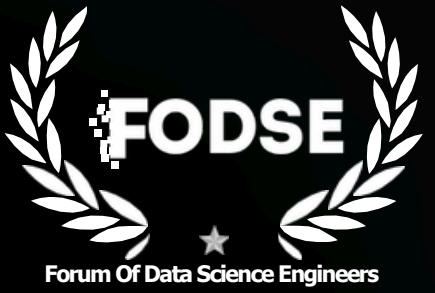
Traffic Analytics & Consulting Firms: Provide data insights for clients or city projects.

- **Technology & AI Companies**

Autonomous Vehicle Developers: Use traffic data for training AI models.

Universities & Research Labs: Studying transportation systems, autonomous vehicles, and AI-based traffic modeling.

TEAM DETAILS



NAME	CONTACT	COLLEGE	ROLE	DEPT & YEAR
Dhanethiren P	6380923883	SVCE	Lead	CSE - III
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